

From Optical Symbol Recognition (OSR) of Point Clouds to the Substation Information Model

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Abstract—Laser scanning today represents one of the most sophisticated geodetic metering methods. This method is becoming more relevant in all substations areas. The challenge is to convert the raw point clouds into smart substation information models. This paper describes how a software provides new tools to radically improve the workflow and the time needed to convert point clouds into intelligent 3D substation models - based on Building Information Modeling (BIM) technology. This then allows experts to take advantage of modern substation engineering methods and calculations.

Index Terms—3D, BIM, point clouds, OSR, substation.

I. INTRODUCTION

Laser scanning, as a method used to capture surfaces of objects and landscapes, is becoming popular for the capture and visualization - in 3D - of already existing substations. Without laser scanning, the common way to get the information of the built substation, as-built, is for experts to directly take measurements inside of the substation. Resulting problems could be offline times of the substation and measurement errors. In the past, laser scanning technology was very cost intensive but it is now an accessible opportunity to use it to capture the real substation and get precise measurements with point clouds - e.g. for brownfield substation projects (extensions, refurbishment, rebuild and retrofit).

Modern laser scanning technologies like Light Detection and Ranging (LIDAR) have the capability to scan 3D inventory with high accuracy. Contemporary software solutions then provide the ability for visualization and editing of these point clouds, obtaining various point densities for different types of objects - independent of their geometric topologies [1]–[3]. It is also possible to obtain the point clouds through image-based technology [4], [5]. LIDAR technology is used in this paper. An example of a point cloud visualization is given in Fig. 1.

The problem is that point clouds only represent a 3D model based on points which means that it is impossible to print this model in technical drawings or to use in further construction processes. The challenge is now how to get a smart BIM substation model - based on point clouds - in order to perform calculations on this model. The current methods to convert point clouds do not result in an intelligent substation model. Due to this, a new method - the Optical Symbol Recognition (OSR) - was developed:

Optical Symbol Recognition (OSR): This newly developed technology assists in the complex and tedious task of converting point clouds into an intelligent 3D model with intelligent 3D symbols, within a substation Building Information Modeling (BIM) based software [7]–[9]. This method is described and presented in this paper.

Examples of conventional methods that are used to convert point clouds into plain 3D models are: 1) Manual conversion: A manual measurement of the point cloud of each object in the Computer Aided Design (CAD) environment is necessary (each electrical device, steel, foundation etc.). Afterwards, defined blocks can be used or newly drafted devices will be placed to get a 3D substation model. This is a time consuming method resulting in a model containing only 2D or 3D geometries without intelligence. 2) Shape recognition: Today, automatic shape recognition with specialized software for point clouds is common and converts the point clouds into 3D solids [6]. This method needs a very high density (resolution) and high level scanning methodology to avoid missing shapes and does not offer an intelligent model as a result.

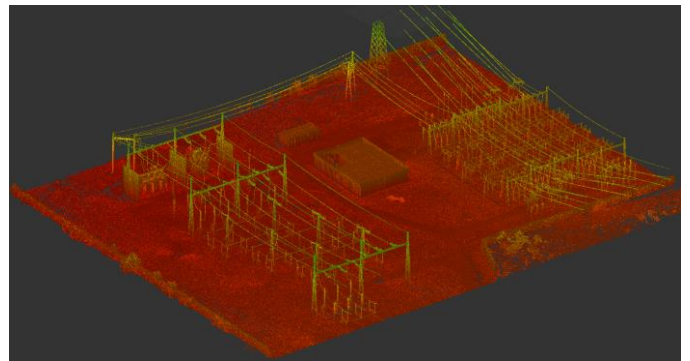


Fig. 1. Point cloud model of a substation

In section II, laser scanning, data collection and processing are introduced. Section III describes the substation BIM software providing the OSR method. Section IV presents the newly developed OSR technology process. Section V then describes further benefits occurring from the use of the OSR technology. Finally, the conclusion and future work are presented.

II. LASER SCANNING OF THE SUBSTATION

Laser scanning represents a collection of spatial data using laser beams where the coordinates of points are determined based on the measured direction and distance. The data of these coordinates are processed to get points and finally a set of those points is called a point cloud [1]. Laser scanning can be divided into two groups: 1) Static laser scanning and 2) Dynamic laser scanning. When the laser scanner is stationary while recording the surfaces, it is static laser scanning. When the laser scanner is attached to a mobile platform, then it is referred to as dynamic laser scanning [1], [2].

Static lasers are more suitable for recording objects, buildings, sites, etc. [1], [2]. Today, static scanning of substations is more common but the dynamic method is becoming more popular. An example of the potential distribution of static laser scanners placed in a substation are shown in Fig. 2. The laser scanner number 8 is shown in Fig. 3.

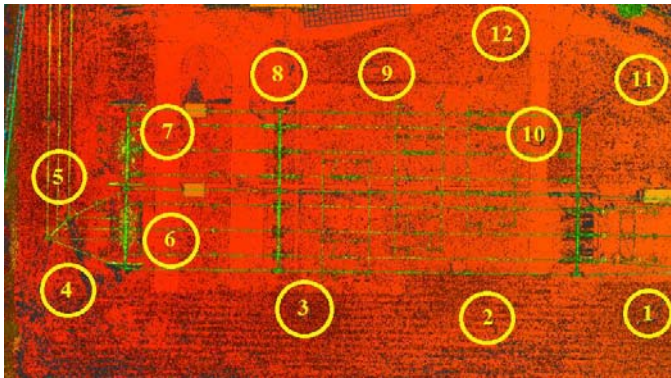


Fig. 2. Example of static laser scanner positions in the substation



Fig. 3. Point cloud image of static laser scanner number 8

New laser scanner technology has become more accessible and can provide precise 3D measurements with a high density of points. Highly detailed 3D data is used and required in various engineering disciplines [3], such as: 1) Virtual planning, analysis of spatial relations between objects, but also between objects and environment; 2) Quality control, supervision and comparison of objects with designs and documentation, especially in complex construction sites and 3) Control of deformations at the sites. Problems that come with the high density of points are file sizes reaching up to

several gigabytes that can be difficult to handle. This includes the need of storage space and powerful machines to handle the large amount of data.

The results of laser scans are files containing all the points captured in the scan process. Each single point contains its coordinate in a defined coordinate system. These files can be read by special point cloud software. Common formats for such files have the following extensions: .asc, .cl3, .clr, .e57, .fls, .fws, .isproj, .las, .pcg, .ptg, .pts, .ptx, .rds, .txt, .xyb, .xyz, .zfs and .zfpj.

After the point cloud data is processed, it is necessary to link the point cloud to a CAD environment. Thereafter, the process to convert to an intelligent substation model can start within a substation BIM software.

III. SUBSTATION BIM SOFTWARE

A detailed description of a substation BIM software used for the research in this paper has already been described in several papers [7]–[9].

This section gives a short introduction to the substation BIM software solution - *primtech* for substation design. The software solution offers users a familiar CAD environment by using the Autodesk AutoCAD family as a graphic editor. According to the BIM philosophy, the experts in the involved disciplines like architects, civil engineers and electrical engineers, can work collaboratively on the same project - greenfield or brownfield - in the same CAD environment [7], [9].

The software provides an extensive library, which contains more than four thousand intelligent 2D and 3D symbols, e.g. electrical devices of different manufacturers, steel structures, foundations, wires and tubes (see example in Fig. 4).

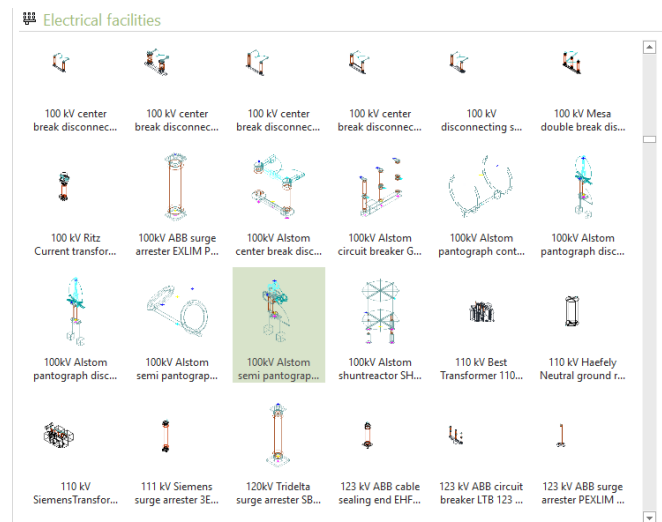


Fig. 4. Example of substation BIM software library

The library has been growing for over 15 years and includes new symbols and symbols for legacy projects. The library contains generic and manufacturer specific symbols. Each symbol (device, steel or foundation, etc.) has a specific geometric signature in the database. The library contains

intelligent symbols with meta data, file links, connection points and defined object behaviors. It offers the possibility to develop the OSR technology to convert point clouds into an intelligent substation information model [7], [8]. The resulting intelligent model is needed to perform calculations in the substation engineering process.

IV. THE OSR PROCESS TO CONVERT A POINT CLOUD INTO A SUBSTATION INFORMATION MODEL

This section describes the Optical Symbol Recognition (OSR) process by describing the method as used within a substation BIM software. As mentioned in section II, after linking the point cloud to the CAD environment, the converting can start. The new OSR technology then converts the point clouds into a substation information model (see OSR process in Fig. 5). OSR was developed within the substation BIM software - *primtech* - as a special tool for symbols recognition in substation point clouds.

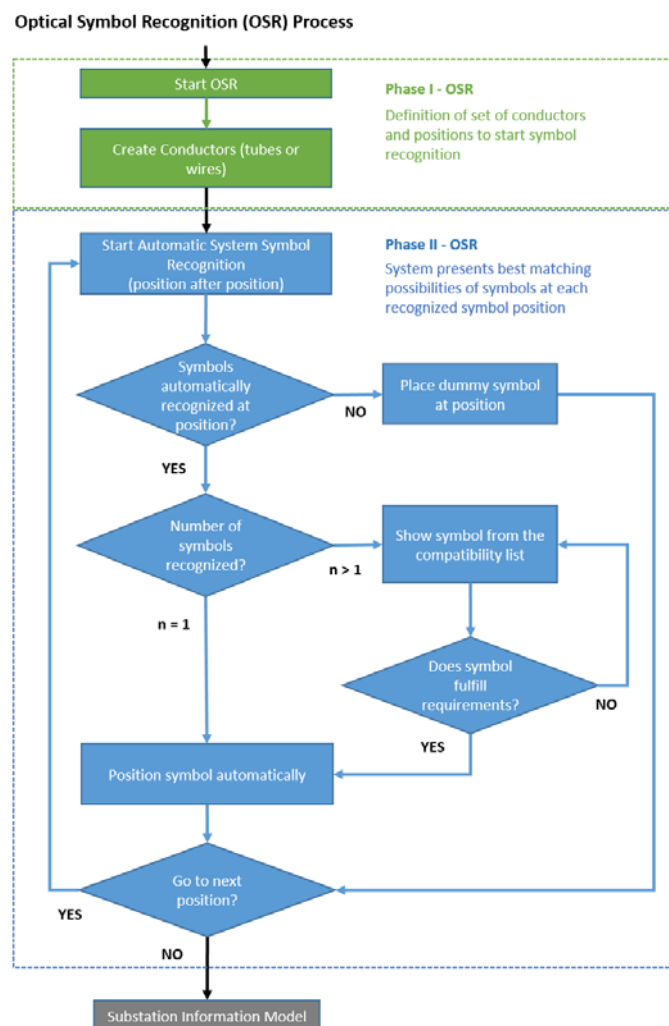


Fig. 5. OSR Process – from point cloud to a substation information model

The OSR process consists of two phases. In phase I the conductors are created. An example of such created conductors is shown in Fig. 6.

The OSR tool provides the opportunity to create a wire by simply selecting three points: 1) First anchor point; 2) Maximum sag point and 3) End anchor point. For the creation of tubes, two or more points can be selected. With this approach, it is possible to create the conductors. Wires and tubes are available in the library for different conductor types according to international standards.

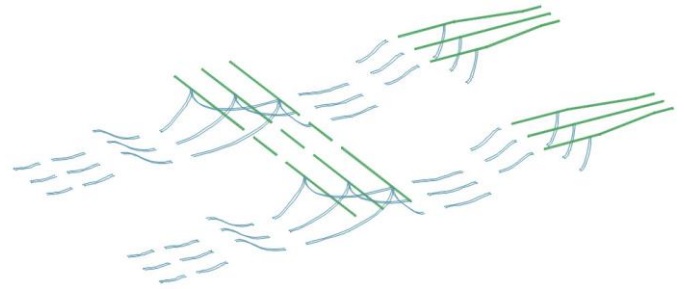


Fig. 6. Example of OSR phase I – created conductors (wires and tubes)

After a set of conductors has been created and placed, the OSR phase II can begin. An OSR heuristic automatically creates symbols as assemblies with electrical device, support structure and foundation heads. An example of this is displayed in Fig. 7.

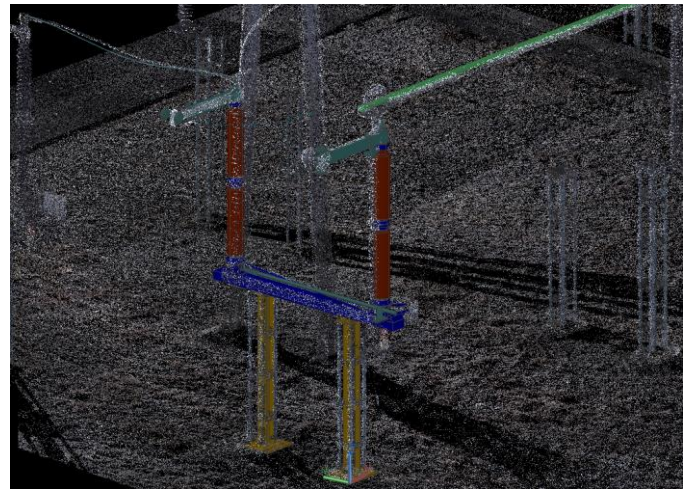


Fig. 7. Recognition of the disconnecting switch and the steel structure

During this phase II, the OSR tool provides a list of compatible library symbols for each symbol type. It is up to the user to choose which symbol should be selected. The user can select the perfect matching symbol or another one in the list and check, e.g. visually, if it fits. A similar workflow applies to all the other symbols like support structures and foundation heads. The system then combines the symbols to create assemblies (electrical device, support structure, foundation head) and places them in the model. Once an assembly has been created, the OSR tool will recognize identical assemblies in the point cloud and propose it at those positions in the substation. The procedure is repeated until no further recognized symbol position is available or the user stops the workflow. For more accurate results, symbols can be filtered by type and voltage levels. Step by step, an intelligent substation model, with the combined intelligent symbols and assemblies, will be created with the OSR technology.

V. BENEFITS OF THE RESULTING SUBSTATION INFORMATION MODEL

The final substation information model in a substation BIM software, converted from a point cloud, is shown in Fig. 8. Once the intelligent BIM model is completed, the software *primtech* offers the possibilities to perform the calculations and use checking tools as well as to create the necessary technical documentation. In an intelligent model, it is then possible to perform e.g. the following calculations according to the international standards: phase checking, clearance calculation, lightning protection calculation, sag and short-circuit effects calculation [8]–[14].

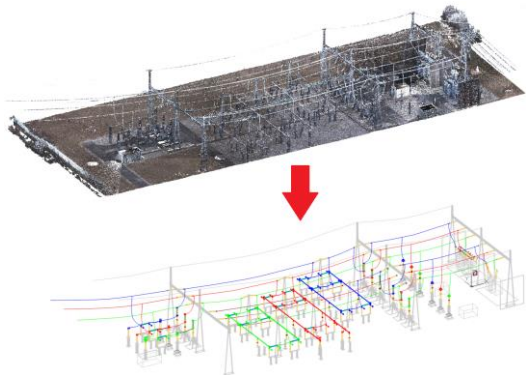


Fig. 8. From point cloud to a substation information model

These calculations are useful in brownfield projects, if e.g. equipment has to be replaced. The clearance calculation can then be used to check the available space for the transport of the equipment. An example of phase checking and clearance calculation running on a point cloud recognition is shown on Fig. 9. In this specific example, a street for the equipment replacement has been placed into the substation. As shown in the figure, the clearance calculation tool displays the safe area where vehicles and maintenance personnel can operate. Also, minimum distance violations between phases 1, 2, 3 and the street were found. With the resulting visualization of the clearances, all necessary precautions can take place for the equipment replacement.



Fig. 9. Clearance calculation in point cloud

CONCLUSION

With the development of this new OSR technology, it is now easier and faster to convert the point clouds into a 3D intelligent substation model. This OSR technology allows a

faster project realization and realistic visualization. It also offers the possibility to run different types of analysis and calculations on the current substation situation and provides a foundation for more accurate and better decision-making process.

The new OSR technology presented in this paper is not fully automatic but drastically improves the conversion of point clouds into intelligent 3D models. Once the substations are converted from point clouds into intelligent substation models, engineers have access to all features already integrated in the substation BIM software, including all available calculations.

In future, the continuing development of this new OSR heuristic technology will implement methods of artificial intelligence (AI) in order to improve recognition and performance even further. The progress will be presented in the future.

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